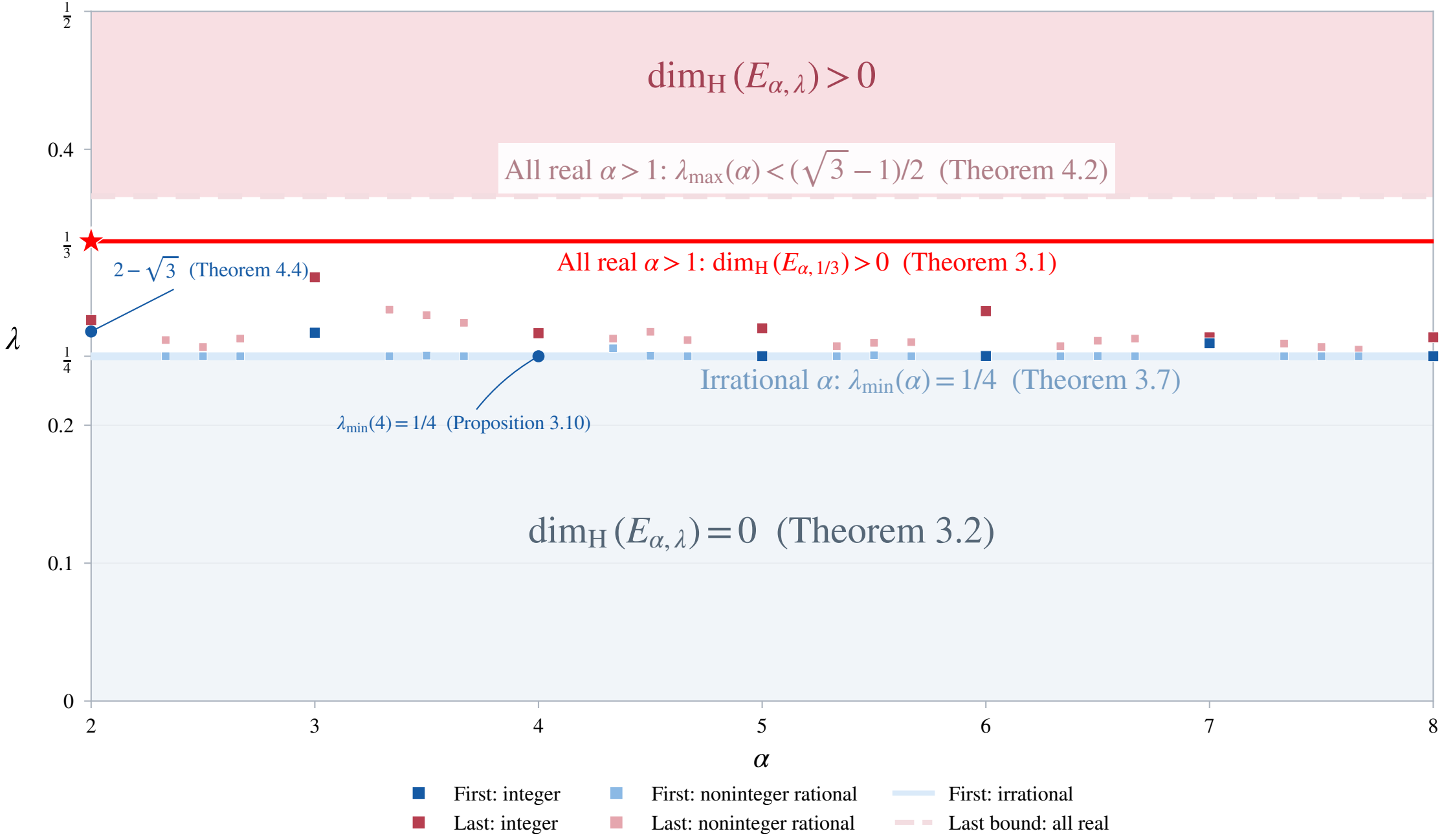
$\dim_H(E_{\alpha,\lambda}) > 0$
All real $\alpha > 1$: $\lambda_{\max}(\alpha) < (\sqrt{3}-1)/2$ (Theorem 4.2)
All real $\alpha > 1$: $\dim_H(E_{\alpha,1/3}) > 0$ (Theorem 3.1)
$2-\sqrt{3}$ (Theorem 4.4)
Irrational α: $\lambda_{\min}(\alpha) = 1/4$ (Theorem 3.7)
$\lambda_{\min}(4) = 1/4$ (Proposition 3.10)
$\dim_H(E_{\alpha,\lambda}) = 0$ (Theorem 3.2)
λ
α
1/2
0.4
1/3
1/4
0.2
0.1
0
2
3
4
5
6
7
8
First: integer
First: noninteger rational
First: irrational
Last: integer
Last: noninteger rational
Last bound: all real
Circles: exact values | Squares: numerical estimates, not rigorous bounds